

MATH802 - ADVANCED FINANCIAL MODELLING AND ANALYTICS

PROJECT

PART 1

The date selected from the January and February S&P 500 data is January two(02JAN2015). The goal is to calculate the VIX for this date of interest. In order to carry out the calculation certain variables need to be calculated initially and applied to the formula. The step by step calculation is provided including the implementation in the SAS through code. The SAS code is constructed in a generic way so that it can be used to calculate the VIX of any date.

STEP BY STEP CALCULATION

- Importing the data and filtering a particular day from the data to carry out the calculation.

```
proc import out = option
datafile= "/home/u63799767/MATH802/option(Jan2015-Feb2015).csv"
dbms="csv";
delimiter=';';
getnames='yes';
run;
proc print data=option(obs = 10);
run;
```

Obs	date	tau	K	call_option_price	S	r	put_option_price
1	01/02/2015	15	1965	98.55	2058.2	0.0015	5.15
2	01/02/2015	15	1970	93.95	2058.2	0.0015	5.55
3	01/02/2015	15	1975	89.4	2058.2	0.0015	6
4	01/02/2015	15	1980	85.05	2058.2	0.0015	6.45
5	01/02/2015	15	1985	80.3	2058.2	0.0015	6.9
6	01/02/2015	15	1990	75.95	2058.2	0.0015	7.5
7	01/02/2015	15	1995	71.6	2058.2	0.0015	8.05
8	01/02/2015	15	2000	67.35	2058.2	0.0015	8.65
9	01/02/2015	15	2005	63.05	2058.2	0.0015	9.35
10	01/02/2015	15	2010	58.75	2058.2	0.0015	10.05

Figure 1

First 10 observations of the data is printed to verify everything is ok.

```
/* Convert Date variable to SAS date format */
data options;
set option;
format Date date9.;
run;
proc print data = options(obs=10);
```

```

run;
/*Filtering data for a particular day */
data jan2;
set options;
where date = '02JAN2015'd;
run;

```

Obs	date	tau	K	call_option_price	S	r	put_option_price
1	02JAN2015	15	1965	98.55	2058.2	0.0015	5.15
2	02JAN2015	15	1970	93.95	2058.2	0.0015	5.55
3	02JAN2015	15	1975	89.4	2058.2	0.0015	6
4	02JAN2015	15	1980	85.05	2058.2	0.0015	6.45
5	02JAN2015	15	1985	80.3	2058.2	0.0015	6.9
6	02JAN2015	15	1990	75.95	2058.2	0.0015	7.5
7	02JAN2015	15	1995	71.6	2058.2	0.0015	8.05
8	02JAN2015	15	2000	67.35	2058.2	0.0015	8.65
9	02JAN2015	15	2005	63.05	2058.2	0.0015	9.35
10	02JAN2015	15	2010	58.75	2058.2	0.0015	10.05

Figure 2

It can be seen that the format of the date is changed to the SAS date format, so that it can be used for the analysis. After changing the date to the right format the data is filtered for a particular day, in this case it is January 2 which is the date of interest.

- After the data is filtered and formatted correctly, the first variable T which is time to maturity is calculated. It can be calculated from tau which is time to option maturity, there will be two tau for one day representing the expiration for near term(minimum value) and next term(maximum value). So there will be two T which will be converted to years by dividing it by 365.

$T1(\text{Near term}) = 15/365 = 0.0410959$

$T2(\text{Next term}) = 35/365 = 0.0958904$

```

/* Calculate T1 and T2 */
T1 = min(tau)/365; /*Near term*/
T2 = max(tau)/365; /*Next term */
/* Initialize Tnew */
Tnew = j(nrow(tau), 1, .);
/* Assign T1 or T2 based on the condition */
do i = 1 to nrow(tau);
if tau[i] = min(tau) then Tnew[i] = T1;
else Tnew[i] = T2;
end;

```

T1	T2
0.0410959	0.0958904

Figure 3

The respective T is stored in the new column called Tnew according to the near and next term.

- There are two risk free interest rate for a day where the minimum value corresponds to near term and the maximum value corresponds to the next term.
 $R1(\text{near term}) = 0.0015$
 $R2(\text{next term}) = 0.0019$
- The forward(F) of the near term(F1) and the next term(F2) needs to be calculated, the condition of the F1 and F2 is given that the difference between the call option price and the put option price should be the lowest for near term and next term. Only the absolute value of the difference is considered. So the difference is calculated and the F is calculated for every strike price and F1 and F2 is decided by the minimum difference in each term.

Difference = Absolute value(call option price – Put option price)

Forward index price(F) = strike price(K) + $(e^{(R \cdot T)}) \cdot (\text{Call price} - \text{Put price})$

R is the interest rate , R1 and R2 for near and next term respectively.

T is the time to maturity , T1 and T2 for near and next term respectively.

CODE:

```
/* Calculate the absolute difference */
Difference = abs(call_option_price - put_option_price);
/* Initialize matrix to store F */
F = j(nrow(k), 1, .);
/* Calculate F for every value of k */
e = exp(1);
do i = 1 to nrow(k);
  F[i] = k[i] + (e##(r[i] * Tnew[i])) * (call_option_price[i] - put_option_price[i]);
end;
/* Find the index where Difference is minimum and tau is minimum and maximum */
/* Locate indices where tau = min(tau) and tau = max(tau) */
idx_tau_min = loc(tau=min(tau));/*Near term*/
idx_tau_max = loc(tau=max(tau));/*Next term*/
min_diff1 = min(Difference[idx_tau_min]);
```

```

idx_min_diff1 = loc(Difference = min_diff1 & tau = min(tau));
min_diff2 = min(difference[idx_tau_max]);
idx_min_diff2 = loc(difference = min_diff2 & tau = max(tau));
/* Extract the corresponding F1 value */
F1 = F[idx_min_diff1];/*Near term*/
F2 = F[idx_min_diff2];/*Next term*/

```

F1	F2
2058.1999	2056.8503

Figure 4

These are the F1 and F2 calculated in the SAS. First difference is calculated and absolute value is considered, then a empty matrix is created to store the value of the F. Then F is calculated for each value of the K .After F is calculated then identifies the indices where the Difference vector has its minimum values corresponding to the minimum and maximum values of tau, denoted as idx_min_diff1 and idx_min_diff2. The values of F at these indices, F1 and F2, are extracted to represent the near-term and next-term conditions respectively.

- Both F1 and F2 will be used to calculate the at the money strike price for near and the next term. The strike price immediately below the forward index level is the at the money strike price for near and next term respectively.

```

/* Determine K0 for near-term (K01) and next-term (K02) options */
K01 = .;
K02 = .;
do i = 1 to nrow(k);
if k[i] <= F1 then K01 = k[i];
if k[i] <= F2 then K02 = k[i];
end;

```

K01	K02
2055	2055

Figure 5

2055 is the at the money strike price for both near term and the next term respectively as they are the strike prices immediately below the forward index price F1 and F2.

- Midquote price for every strike price should be calculated, midquote price can be considered as the average of the call option price and the put option price.
Midquote = (Call option price + Put option price)/2

CODE:

```
/* Calculate the midquote */
midquote = (call_option_price + put_option_price)/2;
```

- The volatility for the near term and the next term should be calculated, in order to calculate the volatility two parts of the volatility is calculated separately and applied to the formula.

Near term volatility:

$$(\sigma_1^2) = (2/T_1) * \sum (\Delta K_i / K_i^2) * (e^{(R_1 * T_1)}) * Q(K_i) - (1/T_1) * [(F_1/K_0) - 1]^2$$

Next term volatility:

$$(\sigma_2^2) = (2/T_2) * \sum (\Delta K_i / K_i^2) * (e^{(R_2 * T_2)}) * Q(K_i) - (1/T_2) * [(F_2/K_0) - 1]^2$$

Contribution by the strike is calculated initially and the total sum needs to be calculated, the equation which is left to the subtraction is considered as part 1 and other is part 2 for easy understanding.

Contribution by strike:

$$(\Delta K_i / K_i^2) * (e^{(R_2 * T_2)}) * Q(K_i)$$

This equation gives the contribution of the strike, it is calculated by applying this formula in the SAS. In order to carry out this calculation delta K should be calculated, this is calculated for near and next term. The first and last value in each term has different formula compared to the values in the middle.

First and last value = (K – adjacent K)

Middle values = Delta_ki => (K(i + 1) – K(i-1))/2

CODE DELTA_K:

```
/* Initialize matrix to store ΔKi */
Delta_K = j(nrow(k), 1, .);
/* Locate indices where tau = min(tau) and tau = max(tau) */
idx_tau_min = loc(tau=min(tau));/*Near term*/
idx_tau_max = loc(tau=max(tau));/*Next term*/
```

```

/* Calculate  $\Delta K_i$  for tau = min(tau) */
do i = 2 to ncol(idx_tau_min) - 1;
Delta_K[idx_tau_min[i]] = (k[idx_tau_min[i]+1] - k[idx_tau_min[i]-1]) / 2;
end;
Delta_K[idx_tau_min[1]] = k[idx_tau_min[2]] - k[idx_tau_min[1]];
Delta_K[idx_tau_min[ncol(idx_tau_min)]] = k[idx_tau_min[ncol(idx_tau_min)]] -
k[idx_tau_min[ncol(idx_tau_min)-1]];
/* Calculate  $\Delta K_i$  for tau = max(tau) */
do i = 2 to ncol(idx_tau_max) - 1;
Delta_K[idx_tau_max[i]] = (k[idx_tau_max[i]+1] - k[idx_tau_max[i]-1]) / 2;
end;
Delta_K[idx_tau_max[1]] = k[idx_tau_max[2]] - k[idx_tau_max[1]];
Delta_K[idx_tau_max[ncol(idx_tau_max)]] = k[idx_tau_max[ncol(idx_tau_max)]] -
k[idx_tau_max[ncol(idx_tau_max)-1]];

```

This calculates the changes (ΔK) in the matrix k at specific indices where the matrix τ reaches its minimum and maximum values. First, it initializes a matrix ΔK with placeholder values. It then locates the indices of the minimum and maximum values of τ , storing them in idx_tau_min and idx_tau_max respectively. For each identified index (excluding the first and last) where τ is at its minimum, it computes the centered difference of the corresponding k values and stores the result in ΔK . For the first and last indices, it calculates simple forward and backward differences respectively. The same procedure is repeated for the indices where τ is at its maximum. This effectively captures the changes in k around points of interest in τ .

After the ΔK is calculated the contribution of the strike can be calculated for each value of K .

CODE CONTRIBUTION BY STRIKE:

```
conbystrike = (Delta_K / (k##2)) # (e##(r # Tnew) # midquote);;
```

Contribution by strike calculated by dividing each element in ΔK by the square of the corresponding element in k , then multiplying the result element-wise by the exponential of the product of r and T_{new} , and finally by midquote . As everything needed for the calculation of the volatility is calculated the two parts required to calculate the volatility is calculated separately and applied to the formula at the end

Part 1: $(2/T) * \sum (\Delta K_i / K_i^2) * (e^{(R*T)}) * Q(K_i)$

Part 2: $(1/T) * [(F/K_0) - 1]^2$

Part 1 and 2 for both near and next term is calculated which is used to calculate the volatility for near and next term. Part 1 is calculated by dividing 2 by the respective T and multiplying it by the sum of the contribution by strike, while part 2 is calculated by dividing 1 by respective t and multiplying it by the term which is calculated by dividing forward index price by the K0 and subtracting it by 1 and raising the whole term to the square.

CODE:

```
p1near = (2/T1)*sum(conbystrike[loc(tau=min(tau))]);
p1next = (2/T2)*sum(conbystrike[loc(tau=max(tau))]);
p2near = (1/T1) * ((F1/K01 - 1)**2);
p2next = (1/T2) * ((F2/K02 - 1)**2);
```

After calculating the part 1 and part 2 volatility can be calculated by calculating the difference between part 1 and part 2.

CODE:

```
sigma1 = p1near - p2near; /* VOLATILITY FOR NEAR TERM */

sigma2 = p1next - p2next; /*VOLATILITY FOR NEXT TERM */
/* Initialize sigma */
sigma = j(nrow(tau), 1, .);
/* Assign T1 or T2 based on the condition */
do i = 1 to nrow(tau);
if tau[i] = min(tau) then sigma[i] = sigma1;
else sigma[i] = sigma2;
end;
```

p1near	p1next	p2near	p2next	sigma1	sigma2
0.0548752	0.038303	0.000059	8.4548E-6	0.0548162	0.0382946

Figure 6

These are the respective values calculated according to the near and next term, the sigma 1 and sigma2 values are stored in a column named sigma according to the near and next term.

- The 30 day weighted average variance of the near and the next term is calculated, which can be used to calculate the VIX.

$$WA = \{ T1 * \sigma_1^2 * [(NT2 - N30)/(NT2 - NT1)] + T2 * \sigma_2^2 * [(N30 - NT1)/(NT2 - NT1)] \} * N365 / N30$$

NT1 – 15 (Number of days to settlement of the near term option)

NT2 – 35 (Number of days to settlement of the next term option)

N30 – 30

N365 – 365 (Number of days in a year)

CODE:

N1 = min(tau);

N2 = max(tau);

N30 = 30;

N365 = 365;

$$wa = ((T1 * \sigma_1^2 * ((N2 - N30) / (N2 - N1))) + (T2 * \sigma_2^2 * ((N30 - N1) / (N2 - N1)))) * (N365 / N30);$$

wa
0.0403598

Figure 7

This is the weighted average calculated using the SAS.

- Finally the VIX is calculated, it is calculated by multiplying the square root of the weighted average by 100.

$$VIX = 100 * \text{sqrt}(\text{weighted average}) \Rightarrow 100 * \text{sqrt}(0.0403598) \Rightarrow 20.089$$

CODE:

vix = 100*sqrt(wa);

vix
20.089742

Figure 8

VIX for the day January 2 is calculated to be 20.089742.

COMPLETE CODE:

```
proc import out = option
datafile= "/home/u63799767/MATH802/option(Jan2015-Feb2015).csv"
dbms="csv";
delimiter=';';
getnames='yes';
run;

proc print data=option(obs = 10);
run;

/* Convert Date variable to SAS date format */
data options;
set option;
format Date date9.;
run;

proc print data = options(obs=10);
run;

/*Filtering data for a particular day */
data jan2;
set options;
where date = '02JAN2015'd;
run;

proc print data = jan2(obs=10);
run;

proc iml;
start main;
use jan2;
read all var {date tau k call_option_price s r put_option_price};

/* Calculate T1 and T2 */
```

```

T1 = min(tau)/365;/*Near term*/
T2 = max(tau)/365;/*Next term */
/* Initialize Tnew */
Tnew = j(nrow(tau), 1, .);
/* Assign T1 or T2 based on the condition */
do i = 1 to nrow(tau);
if tau[i] = min(tau) then Tnew[i] = T1;
else Tnew[i] = T2;
end;
/* Initialize matrix to store  $\Delta K_i$  */
Delta_K = j(nrow(k), 1, .);
/* Locate indices where tau = min(tau) and tau = max(tau) */
idx_tau_min = loc(tau=min(tau));/*Near term*/
idx_tau_max = loc(tau=max(tau));/*Next term*/
/* Calculate  $\Delta K_i$  for tau = min(tau) */
do i = 2 to ncol(idx_tau_min) - 1;
Delta_K[idx_tau_min[i]] = (k[idx_tau_min[i]+1] - k[idx_tau_min[i]-1]) / 2;
end;
Delta_K[idx_tau_min[1]] = k[idx_tau_min[2]] - k[idx_tau_min[1]];
Delta_K[idx_tau_min[ncol(idx_tau_min)]] = k[idx_tau_min[ncol(idx_tau_min)]] -
k[idx_tau_min[ncol(idx_tau_min)-1]];
/* Calculate  $\Delta K_i$  for tau = max(tau) */
do i = 2 to ncol(idx_tau_max) - 1;
Delta_K[idx_tau_max[i]] = (k[idx_tau_max[i]+1] - k[idx_tau_max[i]-1]) / 2;
end;
Delta_K[idx_tau_max[1]] = k[idx_tau_max[2]] - k[idx_tau_max[1]];
Delta_K[idx_tau_max[ncol(idx_tau_max)]] = k[idx_tau_max[ncol(idx_tau_max)]] -
k[idx_tau_max[ncol(idx_tau_max)-1]];
/* Calculate the absolute difference */

```

```

Difference = abs(call_option_price - put_option_price);

/* Calculate the midquote */
midquote = (call_option_price + put_option_price)/2;

/* Initialize matrix to store F */
F = j(nrow(k), 1, .);

/* Calculate F for every value of k */
e = exp(1);
do i = 1 to nrow(k);
F[i] = k[i] + (e##(r[i] * Tnew[i])) * (call_option_price[i] - put_option_price[i]);
end;

/* Find the index where Difference is minimum and tau is minimum */
min_diff1 = min(Difference[idx_tau_min]);
idx_min_diff1 = loc(Difference = min_diff1 & tau = min(tau));
min_diff2 = min(difference[idx_tau_max]);
idx_min_diff2 = loc(difference = min_diff2 & tau = max(tau));

/* Extract the corresponding F1 value */
F1 = F[idx_min_diff1];/*Near term*/
F2 = F[idx_min_diff2];/*Next term*/

/* Determine K0 for near-term (K01) and next-term (K02) options */
K01 = .;
K02 = .;

do i = 1 to nrow(k);
if k[i] <= F1 then K01 = k[i];
if k[i] <= F2 then K02 = k[i];
end;

conbystrike = (Delta_K / (k##2)) # (e##(r # Tnew) # midquote);
p1near = (2/T1)*sum(conbystrike[loc(tau=min(tau))]);
p1next = (2/T2)*sum(conbystrike[loc(tau=max(tau))]);

```

```

p2near = (1/T1) * ((F1/K01 - 1)**2);
p2next = (1/T2) * ((F2/K02 - 1)**2);
sigma1 = p1near - p2near; /* VOLATILITY FOR NEAR TERM*/
sigma2 = p1next - p2next; /*VOLATILITY FOR NEXT TERM*/

/* Initialize sigma */
sigma = j(nrow(tau), 1, .);

/* Assign T1 or T2 based on the condition */
do i = 1 to nrow(tau);
if tau[i] = min(tau) then sigma[i] = sigma1;
else sigma[i] = sigma2;
end;

N1 = min(tau);
N2 = max(tau);
N30 = 30;
N365 = 365;
wa = ((T1*sigma1*((N2-N30)/(N2-N1)))+(T2*sigma2*((N30-N1)/(N2-N1))))*(N365/N30);
vix = 100*sqrt(wa);

print T1 T2 F1 F2 K01 K02 p1near p1next p2near p2next sigma1 sigma2 wa vix;

/* Create jan2new dataset with the calculated values */

create jan2new var {date tau k call_option_price s r put_option_price Difference
midquote Tnew Delta_K conbystrike sigma};

append;

close jan2new;

finish main;

run;

quit;

data jan2new; /* Convert Date variable to SAS date format */

set jan2new;

```

```
format Date date9.;
run;

proc print data = jan2new (obs=10);
run;
```

T1	T2	F1	F2	K01	K02	p1near	p1next	p2near	p2next	sigma1	sigma2	wa
0.0410959	0.0958904	2058.1999	2058.8503	2055	2055	0.0548752	0.038303	0.000059	8.4548E-6	0.0548162	0.0382946	0.0403598

vix
20.089742

Obs	DATE	TAU	K	CALL_OPTION_PRICE	S	R	PUT_OPTION_PRICE	DIFFERENCE	MIDQUOTE	TNEW	DELTA_K	CONBYSTRIKE	SIGMA
1	02JAN2015	15	1965	98.55	2058.2	.0015	5.15	93.40	51.850	0.041096	5	.000067146	0.054816
2	02JAN2015	15	1970	93.95	2058.2	.0015	5.55	88.40	49.750	0.041096	5	.000064100	0.054816
3	02JAN2015	15	1975	89.40	2058.2	.0015	6.00	83.40	47.700	0.041096	5	.000061148	0.054816
4	02JAN2015	15	1980	85.05	2058.2	.0015	6.45	78.60	45.750	0.041096	5	.000058352	0.054816
5	02JAN2015	15	1985	80.30	2058.2	.0015	6.90	73.40	43.600	0.041096	5	.000055330	0.054816
6	02JAN2015	15	1990	75.95	2058.2	.0015	7.50	68.45	41.725	0.041096	5	.000052685	0.054816
7	02JAN2015	15	1995	71.60	2058.2	.0015	8.05	63.55	39.825	0.041096	5	.000050034	0.054816
8	02JAN2015	15	2000	67.35	2058.2	.0015	8.65	58.70	38.000	0.041096	5	.000047503	0.054816
9	02JAN2015	15	2005	63.05	2058.2	.0015	9.35	53.70	36.200	0.041096	5	.000045027	0.054816
10	02JAN2015	15	2010	58.75	2058.2	.0015	10.05	48.70	34.400	0.041096	5	.000042576	0.054816

Figure 9

This is the output from the entire code used to calculate the VIX of that particular day, this can be used to calculate the VIX of any date as it is in generic format. Some of the values calculated during the step by step calculation is stored in a new dataset along with the variables from initial dataset which will be useful in other calculations.

PART 2

IMPLIED VOLATILITY

In this part implied volatility is calculated for the chosen date of interest, which is January 2 in this case. It is calculated for near and next term as both will have a strike price which satisfies the condition of being closest to the underlying market price which is 2058.2 in this case. The new dataset from the previous part is used here which additionally contain the calculated variables apart from the original data. It is coded in a generic way so that when a data for particular day is used , it automatically calculates the implied volatility with needing to input the values, there are two codes separately for near and next term.

CODE:

NEAR TERM

PROC IML;

start main;

use jan2new;

read all var{date tau k call_option_price s r put_option_price Difference midquote Tnew
Delta_K conbystrike sigma};

/* Find K closest to S0 */

idx = loc((abs(k - S) = min(abs(k - S))) & (tau = min(tau)));

/* Extracting values corresponding to the closest strike price and minimum tau */

K = k[idx];

T = Tnew[idx];

r = r[idx];

sigma = sigma[idx];

cop = call_option_price[idx];

S0 = s[idx];

pi = constant("pi");

total_run=3; /* 3 steps */

do i = 1 to total_run;

d1=(log(S0/K)+(r+sigma**2/2)*T)/(sigma*sqrt(T));

d2=d1-sigma*sqrt(T);

c=S0*cdf("Normal",d1)-K*(exp(-r*T)*cdf("Normal",d2));

vega=S0*1/sqrt(2*pi)*exp(-d1**2/2)*sqrt(T);

g=c-cop;

sigma=sigma-g/vega; /* Implied volatility*/

print k T r cop s0 d1 d2 c vega g sigma;

end;

finish main;

```
run;
```

```
quit;
```

k	T	r	cop	S0	d1	d2	c	vega	g	sigma
2060	0.0410959	0.0015	20.85	2058.2	-0.067562	-0.078675	8.3159802	166.07552	-12.53402	0.1302881

k	T	r	cop	S0	d1	d2	c	vega	g	sigma
2060	0.0410959	0.0015	20.85	2058.2	-0.017557	-0.043969	20.869036	166.42934	0.0190365	0.1301737

k	T	r	cop	S0	d1	d2	c	vega	g	sigma
2060	0.0410959	0.0015	20.85	2058.2	-0.017596	-0.043985	20.85	166.42923	6.4583E-9	0.1301737

Figure 10

IML is used for the calculation. This is the code for the near term where the tau is minimum, initially the strike price which is closest to the underlying price is located. Then values like time to expiration, risk free interest rate, sigma(volatility) which was calculated in the previous part and call option price with respect to the closest strike price and the underlying price is extracted. It then initializes a constant pi and sets the total number of iterations to 3. In each iteration, it calculates the parameters d1 and d2, computes the theoretical call option price c, calculates vega (the sensitivity of the option price to volatility), and adjusts sigma using Newton-Raphson iteration to get the implied volatility of the near term. The value from the third iteration where the decimal places are stabilised without any changes are considered as the implied volatility, therefore the implied volatility for the near term on January 2 with underlying market price of 2058.2 is 0.1301737.

NEXT TERM

```
PROC IML;
```

```
start main;
```

```
use jan2new;
```

```
read all var{date tau k call_option_price s r put_option_price Difference midquote Tnew  
Delta_K conbystrike sigma};
```

```
/* Find K closest to S0 */
```

```
idx = loc((abs(k - S) = min(abs(k - S))) & (tau = max(tau)));
```

```
/* Extracting values corresponding to the closest strike price and maximum tau */
```

```
K = k[idx];
```

```

T = Tnew[idx];
r = r[idx];
sigma = sigma[idx];
cop = call_option_price[idx];
S0 = s[idx];
pi = constant("pi");
total_run=3; /* 3 steps */
do i = 1 to total_run;
d1=(log(S0/K)+(r+sigma**2/2)*T)/(sigma*sqrt(T));
d2=d1-sigma*sqrt(T);
c=S0*cdf("Normal",d1)-K*(exp(-r*T)*cdf("Normal",d2));
vega=S0*1/sqrt(2*pi)*exp(-d1**2/2)*sqrt(T);
g=c-cop;
sigma=sigma-g/vega; /* Implied volatility*/
print k T r cop s0 d1 d2 c vega g sigma;
end;
finish main;
run;
quit;

```

k	T	r	cop	S0	d1	d2	c	vega	g	sigma
2060	0.0958904	0.0019	34.1	2058.2	-0.052424	-0.064283	9.0444691	253.91505	-25.05553	0.1369714

k	T	r	cop	S0	d1	d2	c	vega	g	sigma
2060	0.0958904	0.0019	34.1	2058.2	0.0048929	-0.037522	34.12864	254.26116	0.0286402	0.1368587

k	T	r	cop	S0	d1	d2	c	vega	g	sigma
2060	0.0958904	0.0019	34.1	2058.2	0.004862	-0.037518	34.1	254.2612	-2.158E-9	0.1368587

Figure 11

IML is used for the calculation. This is the code for the next term where the tau is maximum(35), initially the strike price which is closest to the underlying price is located. Then values like time to expiration, risk free interest rate, sigma(volatility) which was calculated in the previous part and call option price with respect to the closest strike price and the underlying price is extracted. It then initializes a constant pi and sets the total number of iterations to 3. In each iteration, it calculates the parameters d1 and d2, computes the theoretical call option price c, calculates vega (the sensitivity of the option price to volatility), and adjusts sigma using Newton-Raphson iteration to get the implied volatility of the next term. The value from the third iteration where the decimal places are stabilised without any changes are considered as the implied volatility, therefore the implied volatility for the next term on January 2 with underlying market price of 2058.2 is 0.1368587.

PART 3

The VIX, or Volatility Index, is like a barometer for the stock market's uncertainty levels. When it's low, it suggests investors are feeling confident and things are relatively stable. But when it's high, it indicates a lot of nervousness and turbulence in the market. So, it's a handy tool for gauging overall market sentiment and potential risks. Just like the VIX measures overall market volatility, near and next term implied volatility gauge how much movement traders expect in specific stocks or options over a short period. Near-term implied volatility reflects expectations for the immediate future, while next term looks a bit further ahead. High levels suggest anticipated price swings, while low levels indicate expectations for stability. So, they're useful for traders to assess potential risks and opportunities in specific stocks or options within a shorter time frame.

Repeating the methods in part 1 and part 2 five more VIX and the implied volatilities are calculated for 5 random dates which are JAN15, JAN30 , FEB02, FEB17 and FEB18. The VIX and the implied volatilities calculated for these dates are tabulated below.

DATE	VIX	NEAR TERM IMPLIED VOLATILITY	NEXT TERM IMPLIED VOLATILITY
02JAN2015	20.089742	0.1301737	0.1368587
15JAN2015	24.994294	0.2157911	0.1774996
30JAN2015	25.189093	0.1882955	0.1703417
02FEB2015	22.507105	0.1446644	0.1487932
17FEB2015	19.715015	0.1078536	0.120316
26FEB2015	18.530757	0.0929367	0.1100007

These values are stored in a dataset and plotted in SAS for the better understanding of the trend between VIX and the implied volatility of the near term and the next term. Each of the implied volatility is plotted against the volatility index with VIX on the x axis and implied volatilities in the y axis.

CODE:

DATA part3;

```
/* Define the variables in the dataset */
```

```
INPUT DATE : $9. VIX NEARTERMIMPLIEDVOLATILITY NEXTTERMIMPLIEDVOLATILITY;
```

DATALINES;

```
02JAN2015 20.089742 0.1301737 0.1368587
```

```
15JAN2015 24.994294 0.2157911 0.1774996
```

```
30JAN2015 25.189093 0.1882955 0.1703417
```

```
02FEB2015 22.507105 0.1446644 0.1487932
```

```
17FEB2015 19.715015 0.1078536 0.120316
```

```
26FEB2015 18.530757 0.0929367 0.1100007
```

```
;
```

RUN;

```
/* Sort the dataset by VIX */
```

```
PROC SORT DATA=part3;
```

```
BY VIX;
```

RUN;

```
proc sgplot data=part3;
```

```
title 'VIX AND IMPLIED VOLATILITY';
```

```
series x=vix y=neartermimpliedvolatility / legendlabel='Near term  
IV' lineattrs=(color=blue);
```

```
series x=vix y=nexttermimpliedvolatility / legendlabel='Next term IV'  
lineattrs=(color=red);
```

```
xaxis label='VIX';
```

```
yaxis label='IMPLIED VOLATILITY';
```

run;

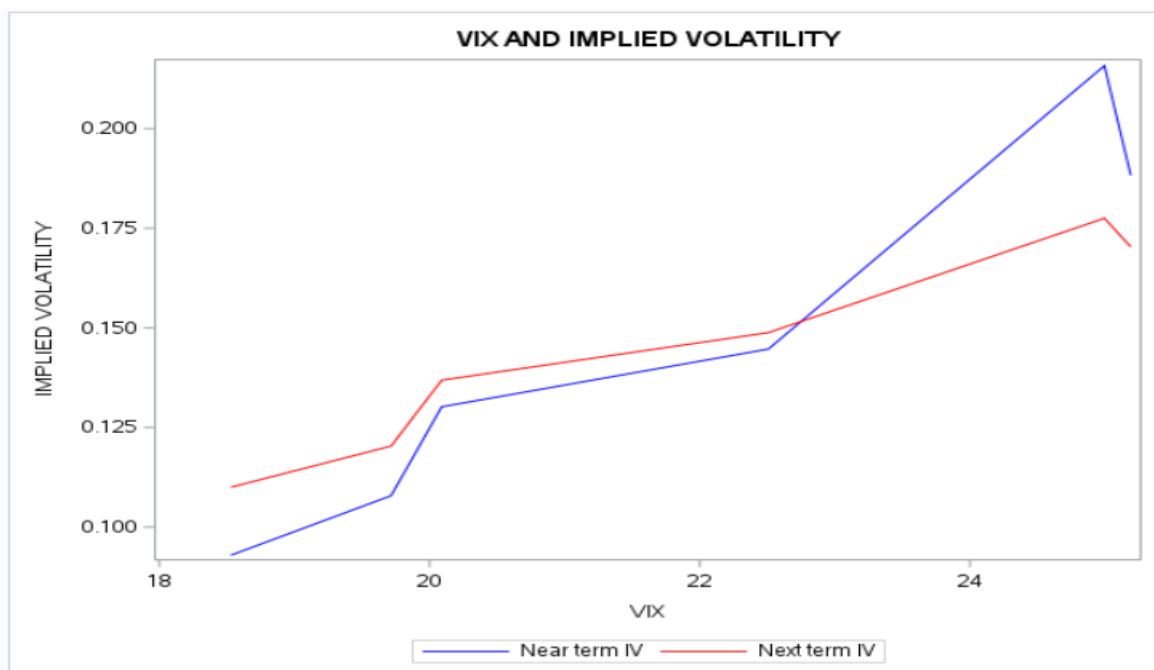


Figure 12

VIX and Implied Volatility Trends

The plot illustrates the relationship between the VIX (Volatility Index) and implied volatility (IV) over different time frames, specifically the near-term and next-term implied volatility. Observing the graph, there is a clear upward trend in both the VIX and IV as the values progress from left to right on the x-axis. The near-term IV, represented by the blue line, exhibits a sharper increase compared to the next-term IV, denoted by the red line. This suggests that as the VIX rises, investors expect higher volatility in the near term more significantly than in the next term. The peaks and troughs in the near term IV indicate a higher sensitivity to immediate market conditions, while the next term IV shows a more gradual and stable increase. It can be noticed that initially the near term implied volatility is lesser than the next term implied volatility, after certain value of VIX this trend changes as near term implied volatility becomes higher than the near term implied volatility.

Differences and Similarities Between VIX and Implied Volatility

The key differences between the two measurements lie in their respective reactions to market conditions. Near-term IV tends to be more reactive to current market news and short-term uncertainties, resulting in sharper fluctuations. This is evident from the steeper ascent and more pronounced changes in the blue line. In contrast, next-term IV represents a more smoothed out response, reflecting longer-term market expectations and exhibiting less volatility in its movement. Despite these differences, both

measurements share a fundamental similarity: they both trend upward with the VIX, indicating that as market uncertainty (represented by the VIX) increases, the expectation of future volatility (both near-term and next-term) also rises.

Financial Interpretation of VIX and Implied Volatility

From a financial perspective, the VIX is often referred to as the "fear gauge" of the market, providing insight into investor sentiment and expectations of market volatility. The implied volatility, on the other hand, represents the market's forecast of a likely movement in an asset's price and is derived from the prices of options. A rising VIX typically signals increasing uncertainty or risk in the market, leading to higher premiums on options as traders anticipate larger price swings. The near-term IV's more aggressive increase compared to the next-term IV suggests that investors are particularly concerned about short-term market instability. This can be due to immediate economic events, earnings reports, or geopolitical tensions. In essence, while both VIX and IV are integral in gauging market sentiment, the near-term IV provides a more immediate snapshot of market fears, whereas the next-term IV offers a more tempered outlook. This differentiation is crucial for traders and investors in strategizing their positions and managing risk accordingly.